

1) $R_f = 0.55 \text{ cm}$ $LHR = 300 \text{ W/cm}$ $K_f = 0.025 \text{ W/cm-K}$
 $E = 200 \text{ GPa}$ $\nu = 0.35$ $\alpha = 10 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$

a) max stress? σ_θ @ $r = R_f$

$$\sigma_\theta = -\sigma^* (1 - 3\eta^2) \quad \sigma^* = \frac{\alpha E \Delta T}{4(1-\nu)}$$

$$\Delta T = T_i - T_s = \frac{LHR}{4\pi K_f} = \frac{300}{4\pi (0.025)} = 955 \text{ K}$$

$$\sigma^* = \frac{(10 \times 10^{-6})(200)(955)}{4(1-0.35)} = 735 \text{ MPa}$$

$$\eta @ r = R_f = 1$$

$$\sigma_\theta = -\sigma^* (1 - 3(1)^2) = 2\sigma^* = \underline{1469 \text{ MPa}}$$

b) η where $\sigma_{\theta r} = \sigma_\theta$ $\sigma_{\theta r} = 175 \text{ MPa}$

$$175 = -735 (1 - 3\eta^2) \quad \eta^2 = 0.413$$

$$\eta = 0.64 \rightarrow \underline{36\% \text{ into the fuel}}$$

2) cladding w/ $p = 15 \text{ MPa}$ $\bar{R}_c = 0.55 \text{ cm}$ $t_c = 0.05 \text{ cm}$

a) Thin

$$\sigma_\theta = \frac{pR}{\delta} = \frac{15(0.55)}{0.05} = \underline{165 \text{ MPa}}$$

$$\sigma_z = \frac{pR}{2\delta} = \frac{15(0.55)}{2(0.05)} = \underline{82.5 \text{ MPa}}$$

$$\sigma_r = \frac{-p}{2} = \frac{-15}{2} = \underline{-7.5 \text{ MPa}}$$

b) Thick, $r = R_c$

$$\sigma_\theta = p \frac{(R_o/r)^2 + 1}{(R_o/r)^2 - 1}$$

$$\sigma_r = -p \frac{(R_o/r)^2 - 1}{(R_o/r)^2 - 1}$$

$$\sigma_z = \frac{p}{(R_o/r)^2 - 1}$$

$$\frac{R_o}{R_c} = \frac{0.55 - 0.05/2}{0.55 - 0.05/2} = 1.095$$

$$\sigma_\theta = 15 \frac{1.095^2 + 1}{1.095^2 - 1} = \underline{165.7 \text{ MPa}}$$

$$\sigma_r = -15 \frac{1.095^2 - 1}{1.095^2 - 1} = \underline{-15 \text{ MPa}}$$

$$\sigma_z = \frac{15}{1.095^2 - 1} = \underline{75.4 \text{ MPa}}$$

$$3) \Delta \delta_g = \bar{R}_c \alpha_c \Delta T_c - R_f \alpha_f \Delta T_f$$

$$R_f = 0.48 \text{ cm} \quad b_g = 0.008 \text{ cm} \quad T_{c0} = 550 \text{ K} \quad t_c = 0.08 \text{ cm} \quad K_f = 0.03 \text{ W/cmK}$$

$$K_c = 0.15 \text{ W/cmK} \quad LHR = 250 \text{ W/cm} \quad \alpha_c = 10 \times 10^{-6} \text{ } ^\circ\text{K}^{-1} \quad \alpha_f = 14 \times 10^{-6} \text{ } ^\circ\text{K}^{-1}$$

$$T_{ref} = 300 \text{ K} \quad K_g = 0.003 \text{ W/cm}$$

$$T_{c1} - T_{c0} = \frac{LHR}{2\pi R_f} \frac{t_c}{K_c} = \frac{250}{2\pi \cdot 0.48} \frac{0.08}{0.15} = 44.2 \text{ K} \quad T_{c1} = 594.2 \text{ K}$$

$$T_g - T_{c1} = \frac{LHR}{2\pi R} \frac{b_g}{K_g} = \frac{250}{2\pi \cdot 0.48} \frac{0.008}{0.003} = 221 \text{ K} \quad T_g = 815.2 \text{ K}$$

$$T_0 - T_g = \frac{LHR}{4\pi K_f} = \frac{250}{4\pi \cdot 0.03} = 663 \text{ K} \quad T_0 = 1478.2 \text{ K}$$

$$\bar{T}_c = \frac{T_{c1} + T_{c0}}{2} = \frac{594.2 + 550}{2} = 572.1 \text{ K}$$

$$\Delta \delta_c = \bar{R}_c \alpha_c \Delta T_c \quad \bar{R}_c = 0.48 + 0.008 + \frac{0.08}{2} = 0.528 \text{ cm}$$

$$= 0.528 (10 \times 10^{-6}) (572.1 - 300) = 1.44 \times 10^{-3} \text{ cm}$$

$$\bar{T}_f = \frac{1478.2 + 815.2}{2} = 1146.7 \text{ K}$$

$$\Delta \delta_f = 0.48 (14 \times 10^{-6}) (1146.7 - 300) = 5.67 \times 10^{-3} \text{ cm}$$

$$\Delta \delta_g = 1.44 \times 10^{-3} - 5.67 \times 10^{-3} = \boxed{-0.00423 \text{ cm}}$$

$$t_g' = 0.008 - 0.00423 = \boxed{0.00377 \text{ cm}}$$

4) $\sigma_r + \sigma_\theta$ @ $r = R_i$:

$\Delta T = 50 \text{ K}$ $\alpha_c = 15 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ $E = 100 \text{ GPa}$ $\nu = 0.34$

$t_c = 0.04 \text{ cm}$ $R_{ic} = 0.55 \text{ cm}$

$$\sigma_r = \frac{\Delta T}{2} \frac{\alpha E}{1-\nu} \left(\frac{r}{R_i} - 1 \right) \left(1 - \frac{R_i}{r} \left(\frac{r}{R_i} - 1 \right) \right)$$

$r = R_i \rightarrow \sigma_r = 0$

$$\sigma_\theta = \frac{\Delta T}{2} \frac{\alpha E}{1-\nu} \left(1 - \frac{2R_i}{r} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$\begin{aligned} \sigma_\theta &= \frac{50}{2} \frac{(15 \times 10^{-6})(100)}{1-0.34} \left(1 - \frac{2(0.55)}{0.04} \left(\frac{0.55}{0.55} - 1 \right) \right) \\ &= \underline{56.8 \text{ MPa}} \end{aligned}$$

5) $\epsilon_0 = \Delta \rho_0 \left(\exp \left(\frac{\Delta h_{0.01}}{C_0 \beta_0} \right) - 1 \right)$

$t = 10 \text{ } \mu\text{s}$

$\dot{\Gamma} = 6 \times 10^{13} \frac{\text{kg}}{\text{cm}^2 \cdot \text{s}}$ $\Delta \rho_0 = 0.015$ $\beta_0 = 5 \text{ mW/kg}$ $\rho = 10.47 \text{ g/cc}$

$\beta = \frac{\dot{\Gamma} t}{N_u}$ $N_u = 10.47 \text{ g/cc} \frac{1 \text{ mol}}{270 \text{ g}} \frac{4.02 \times 10^{-2} \text{ J}}{1 \text{ mol}} = 2.447 \times 10^{-22} \text{ J/cc}$

$= \frac{(6 \times 10^{13})(10 \times 24 \times 60 \times 60)}{2.447 \times 10^{-22}} = 2.0021 \times 10^{22} \text{ J/cc} = \beta \text{ (final)}$

$\beta(\text{final}) = \frac{5}{950} = 0.0053$

$\epsilon_0 = 0.015 \left(\exp \left(\frac{0.0021 \ln 0.01}{(1) 0.0053} \right) - 1 \right)$

$\underline{\epsilon_0 = -0.0126}$

6. (4 pts) Describe both elasticity and plasticity.

Elasticity is a stretching of bonds that results in a temporary, reversible deformation. Plasticity is a permanent deformation due to the breaking of bonds, via slip or twinning.

7. (8 pts) Define strain hardening. What causes strain hardening?

Strain hardening is the propensity of a material to increase its yield stress after undergoing plastic deformation. Dislocations are the main mechanism of plasticity, and plastic deformation both forms additional dislocations and causes them to move. As more dislocations are present, dislocation-dislocation repulsion becomes more prevalent, leading to dislocation pileup on barriers. This pileup makes it harder for dislocations to move, and thus, results in an increase in the yield strength.

8. (3 pts) What three things must all fuel performance codes be able to do?

Model the temperature in the fuel, model the stresses in the cladding, model the evolution of the gap.

9. (6 pts) How do grains form in the sintering process of fuel?

Grains originate from the powder particles. With applied temperature and pressure, the particles come into contact with arbitrary crystalline orientations with respect to one another. Diffusive processes remove void space, leaving a 2-D defect, the grain boundary, between the two original neighboring particles. Occurring on a multi-particle scale, this results in a polycrystalline structure.

10. (8 pts) Describe the concept of microstructure-based (mechanistic) fuel performance modeling and why it is beneficial.

Instead of using empirical burnup-based models, utilize microstructure-based models which take into account the fundamental physical mechanisms, processes, and material properties of the underlying materials. This process should lead to more flexible predictive models of fuel performance that can be applied both where historical experimental data exists, as well as where no data exists.

11. (8 pts) What is meant by microstructure? Given an example of a processing technique and the impact it has on the microstructure and related properties.

Microstructure is loosely defined as what can be observed at 25X. This includes grain structures and morphologies, second phases, large precipitates or gas bubbles, etc. Annealing is a processing technique that involves holding a material at an elevated temperature for an extended period of time. This process allows for the diffusion and annihilation of defects, resulting in larger grained structures with lower defect densities, and thus, materials with typically lower yield stresses and higher ductility.

12. (4 pts) Does diffusion occur more quickly inside the grains, or on the grain boundaries? Why?

On the grain boundaries. This is because there is a non-ideal bonding environment and typically more space in the grain boundaries, which makes diffusive processes easier. Grain boundary diffusion becomes even more dominant at lower temperatures.

13. (6 pts) What is a driving force for grain growth? What can impede grain growth?

Reduction in the total grain boundary energy. Often referred to as curvature-driven grain growth, leads to smaller grains shrinking and larger grains growing. Grain growth can be impeded by obstacles residing at the grain boundaries. This includes precipitates, bubbles, voids, etc.

14. (6 pts) Why does fuel densify? Describe this process.

Fuel densification occurs as a continuation of the sintering process. Fuel is fabricated with less than 100% density, and thus, there are residual pores present in fresh fuel. Under operation, diffusion and fission spikes can combine to result in the destruction or removal of small pores, while larger pores typically remain stable. Thus, a fraction of the original porosity is removed.